

Gustavo Lazzari

gustavotlazzari@gmail.com | +55 41 98773-2772 | linkedin.com/in/glazzari | github.com/GLazzari1428 | Curitiba, PR, Brazil

EDUCATION

Pontifical Catholic University of Paraná | *B.Sc. Computer Science* **2024 – 2028**

- Expected graduation: June 2028 | Operating Systems, Databases, Network Security, Data Structures

Swiss School of Curitiba | *High School & IB Bilingual Diploma* **2017 – 2020**

EXPERIENCE

Court of Justice of Paraná (TJPR) **Curitiba, Brazil**

IT Intern

Aug 2021 – Dec 2022

- Deployed and maintained containerized applications using Docker and Docker Compose on cloud infrastructure (AWS EC2, GCP Compute Engine).
- Implemented CI/CD automation with GitHub Actions; configured Nginx and Apache web servers for application hosting.

PROJECTS

Smart Home IoT Network | *10+ ESP32 Devices / ESPHome / MQTT* **Ongoing**

- Deploy and maintain a network of 10+ ESP32 microcontrollers for smart lighting control, intelligent switches, and ambient LED installations (WLED), fully configured via ESPHome using YAML and Jinja2 templating.
- Integrate all IoT devices with Home Assistant through MQTT for centralized automation rules, scheduling, and real-time monitoring of the home environment.
- Currently building a custom Home Assistant remote controller using salvaged Cherry MX mechanical switches and an ESP32, designed and configured entirely through ESPHome.
- Replaced commercial voice assistants (Alexa) with custom ESP32-based solutions, maintaining full local control without cloud dependencies.

AI-Powered Security & Surveillance | *Frigate NVR / Object Detection* **Ongoing**

- Deployed Frigate NVR for real-time local video processing with AI-driven object detection, integrated with Home Assistant for automated alerting and event-driven actions.
- Configured GPU-accelerated video processing pipeline using Quadro P400 with hardware passthrough in Proxmox VE.

Supporting Server Infrastructure | *Proxmox / Docker / Monitoring* **Ongoing**

- Operate a two-node Proxmox VE cluster (20+ VMs/LXC's) providing the backbone for all IoT services, including MQTT brokers, Home Assistant, Frigate, and monitoring infrastructure.
- Configured secure remote access via WireGuard VPN and Cloudflare Zero Trust, enabling reliable control of home automation from anywhere.
- Maintain observability with Prometheus and Grafana for system metrics, Scrutiny for disk health monitoring, and automated backups with offsite replication.

TECHNICAL SKILLS

IoT & Embedded ESP32, ESPHome (YAML/Jinja2), MQTT, WLED, sensor integration, relay control, microcontroller deployment

Home Automation Home Assistant (advanced automations, integrations), Frigate NVR (AI object detection), Navidrome

Hardware & Electronics Soldering (8+ years daily experience), component-level repair, desoldering, PCB modification, custom enclosure design

Infrastructure Linux Admin (Arch/Debian), Proxmox VE, Docker, Docker Compose, ZFS, systemd

Networking WireGuard, Cloudflare (Tunnels/Zero Trust), Nginx, PiHole, OPNsense, VLAN segmentation

Scripting & Automation Bash, Python, YAML/IaC, GitHub Actions

Languages Portuguese (Native), English (Fluent), German (Intermediate)

CERTIFICATIONS

Cambridge Advanced English (CAE – B2) | Deutsches Sprachdiplom (DSD – B1/B2) | Currently pursuing: AWS Certified Cloud Practitioner (CLF-C02)